

Super network

Delivering food and oxygen to all your hardworking cells and removing their wastes is a 24-hour task and one that requires a superefficient transportation system. This is exactly what your body has—an astonishing network of living tubes called blood vessels that transport the vital supplies. Altogether, about 60,000 miles (100,000 km) of blood vessels fit into your body and keep your cells working.

PERFECT MATCH

You have three types of blood vessels—arteries, veins, and capillaries. Each is perfectly built for its role as a blood transporter. Your arteries, for example, carry oxygen-rich blood from your heart to the rest of your body and can withstand the full force of the high pressure of blood that surges through them. Your veins, however, return oxygen-poor blood back to your heart and carry blood under lower pressure, while your capillaries deliver the fuel and oxygen to individual cells.

► **CAPILLARY** *This is the smallest, the most numerous, and the most fragile of your blood vessels.*

The thin wall of a capillary is just one cell thick.

► **VEIN** *A vein is similar in structure to an artery, but some veins also have valves that prevent blood from flowing backward.*

A valve opens and closes like a trapdoor to keep blood flowing in the right direction.

Inner lining of vein

Elastic layer

Thin muscle layer

Protective coat

► **ARTERY** *Thick muscle and elastic layers allow arteries to withstand the high blood pressure created when the heart beats.*

A smooth inner lining allows blood to flow through easily.

This elastic layer allows the artery to expand and bounce back as blood surges through it.

Tough, thick muscle layer

Protective coat

WOW!

If all your blood vessels were laid out end-to-end, they would wrap around the Earth twice.